

Physiological Waveform Imputation of Missing Data using Convolutional Autoencoders

Daniel Miller, Andrew Ward, Nicholas Bambos
Department of Electrical Engineering
Stanford University, Stanford, CA 94305
Email: {danielrm, atward, bambos}@stanford.edu

David Scheinker
Department of Management Science and Engineering
Stanford University, Stanford, CA 94305
Email: dscheink@stanford.edu

Andrew Shin
Division of Pediatric Cardiology
Stanford University School of Medicine, Stanford, CA 94305
Email: drewshin@stanford.edu

Abstract—Machine Learning has great potential to improve automated real-time patient diagnostics. For the majority of machine learning algorithms, taking advantage of this potential requires a complete dataset with no missing data. In practice, missing values are estimated using a variety of imputation methods in the pre-processing stage. However, with time-series data, and physiological waveforms in particular, imputation can be difficult due to the unique patterns and shapes of each waveform, as well as how these patterns vary between patients, and even for a single patient over longer durations. We demonstrate that deep learning techniques can reconstruct missing data using patient-specific patterns present in the non-missing portions of the waveform. Using convolutional neural network (CNN) autoencoders trained on 288 15-minute samples from each of 138 pediatric patients, we develop a generalizable model to analyze and extract information from arbitrary physiological waveforms, and use this model to develop methods for mid-channel missing time-series imputation. We further show that the autoencoder can be used to compress the dense physiological waveforms to a low-dimensional representational space.

I. INTRODUCTION

Medical waveform data contain a wealth of information about the heart, lung, and brain activity of hospital patients. Modern machine learning techniques are beginning to leverage this time-series information to assist in medical decisions. For example, machine learning and deep learning systems which use electrocardiograms (ECGs) to identify heart arrhythmias [1], [2], or which use electroencephalographs (EEGs) to detect epileptic seizures [3], have been realized in research, and avenues have been provided for these systems to provide in-hospital clinical decision support [4].

The medical waveforms that enable these technologies are often sampled at high frequencies for long time periods, generating a large amount of data, which can be difficult to store. In addition, patient waveforms are often riddled with missing or corrupted data. Managing the storage and processing of these datasets, as well as handling missing time-series data, is an active area of research.

Often, missing ECG or related waveform data in research is handled by discarding the time series which have missing data. The remaining data is usually sufficient for a research study.

An operational system, though, needs to be able to handle missing data in real-time.

As a motivating example, consider a machine learning system which monitors a patient's ECG for heart arrhythmias. The system extracts features from the previous hour of the patient's ECG to determine whether the patient is at risk for heart arrhythmias in the next five minutes. However, ECG leads often fall off or experience similar disruptions. Even disconnecting the lead for a few seconds could affect the machine learning system's performance for the next hour. A data imputation algorithm would be able to fix this problem.

For time-series data, common imputation solutions include simple statistical methods, such as mean or median value imputation, linear regression, or expectation maximization [5], [6], [7], [8], [9]. Other studies have proposed more complex methods for imputing missing medical waveform data, such as using vector autoregressive-based imputation for ECGs [10]. Unfortunately, these techniques are often ineffective, particularly when large adjacent sections of the data are missing.

Machine learning-based methods have been shown to be effective at time-series imputation within machine learning pipelines [11]. Several studies have performed imputation using deep learning on image representations (spectrograms) of waveform data, for both ECG data [12] and EEG data [13].

Although deep learning imputation for medical waveforms is relatively sparsely studied, much work has been done in the imputation and generation of audio waveforms. Autoregressive deep generative networks have been successfully applied to state-of-the-art text-to-speech applications [14], [15], [16]. Variational autoencoders, another unsupervised approach, have been used on audio waveforms to extract low-level features without human supervision [17] and to remove noise and enhance speech [18]. A recent study presented SEGAN, a generative adversarial network which aims to remove noise and enhance speech in audio, for mobile communication and hearing aids [19]. These audio-based studies can be translated and applied to physiological data.

A. Our Contributions

Our primary contribution is a class of convolutional-deconvolutional neural network autoencoders operating on medical waveform data. We describe two different models. The first is for compression: taking a time series as input, compressing it to a small dimensionality, and then reconstructing an estimate of the original signal. The second network is for imputation: taking a time series with a significant continuous section artificially removed, and estimating the removed segment. Such a network would be a valuable tool in both medical and more general time-series machine learning systems. In addition, we introduce a newly collected and expansive dataset of pediatric medical waveforms. The sheer size of the collection enables many future unsupervised learning projects. Although this dataset is not manually labeled, patient waveforms can be linked to additional hospital records, providing labels for future supervised learning.

II. METHODS

A. Study Participants

The dataset used for this study was sampled from 138 randomly chosen historical and fully anonymized patient records. This work was approved by the LPCH Institutional Review Board. We considered only samples which included at least 15 minutes of concurrent data for both the electrocardiogram (ECG) and respiratory (RESP) waveforms. From each patient record, we included 288 15-minute samples, for a total sample count of 39744. Although the model did not significantly overfit the training data, future work will incorporate a larger sample size. This dataset was randomly split on a per-patient basis into a standard partition of 80% training set, a 10% model validation set, and a 10% test set.

B. Model Development

A neural network model can be represented by an iterated sequence of mathematical operations, or “layers,” applied to either all of or a portion of the input data. Each layer is parameterized by some number of weight and bias values. The final layer of the network defines a *loss* or *objective* function which represents how well the network achieves its purpose. Training the network involved repeatedly back-propagating gradient computations through each layer to determine how to adjust the weight values so that the loss function will be minimized. Each time this training process iterates over the entire training dataset is referred to as an *epoch*. Training a network may require dozens or hundreds of epochs to fully minimize the loss function. However, because a neural network might contain millions of parameters, often significantly more than the number of samples in the dataset, various techniques are used to reduce the effects of over-fitting the train data. For this reason, a separate model validation set is used to evaluate each model’s performance per epoch.

Convolutional neural networks (CNNs) are a specific family of architectures designed to take advantage of spatially or temporally correlated data. They consist of a set of filters which are swept, or convolved, over the primary dimensions,

in this case time. CNNs are highly parallelizable compared to other architectures like Recurrent networks (RNNs), allowing for efficient application to the extremely dense physiological waveforms of this problem.

Autoencoders consist of first a leading sequence of layers used to transform, or encode, the input, often compressing it into a lower-dimensional space, followed by an mirrored sequence of layers used to reconstruct the compressed representation into an estimation of the original input. The applications of autoencoder networks include signal de-noising, data compression, and data generation. In Section IV we use the encoder and decoder components of a separately trained autoencoder network to impute missing waveform segments.

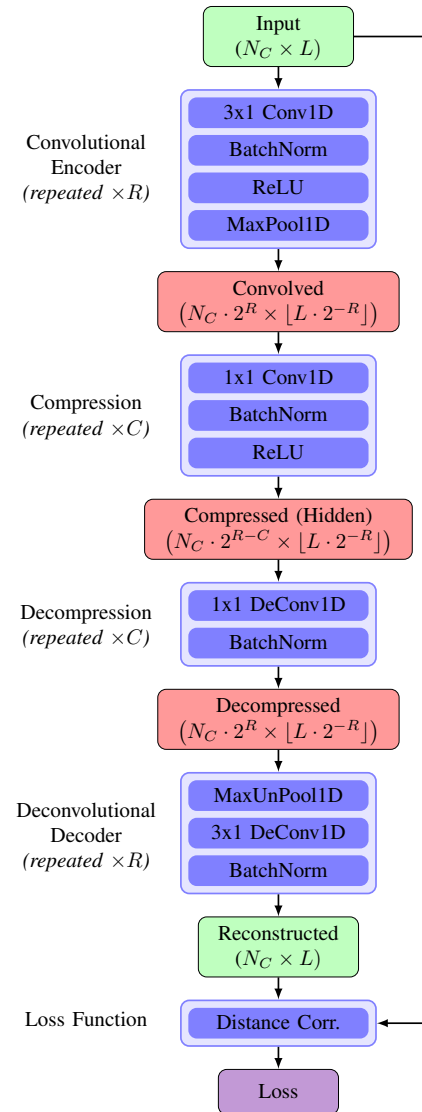


Fig. 1. Convolutional Autoencoder Network. Input waveforms are fed through a sequence of convolutional layers, then a sequence of compression layers, resulting in an encoded hidden state. This hidden state is then reconstructed using a mirrored sequence of decompression and deconvolutional layers, resulting in an approximate reconstruction of the original input waveform.

We construct an autoencoder comprised of 1-dimensional

convolutional layers to analyze the physiological waveforms. The network architecture is shown in Figure 1. This network begins with a set of R identical convolutional encoders used to capture temporal correlations and patterns. Each repetition of the encoder max-pools the time dimension to half its previous size, rounded down, and doubles the number of channels used to represent the information. This is followed by a set of compression layers. Each compression layer uses a 1×1 convolution layer to combine the previous layer's channels into a lower-dimensional representation of half the size. This results in a low-dimensional hidden state with size primarily dependent on the number of compression layers C . The hidden state is then decompressed, and reconstructed to the original size using deconvolutional layers. The output time series is then compared against the input.

C. Evaluation Metric

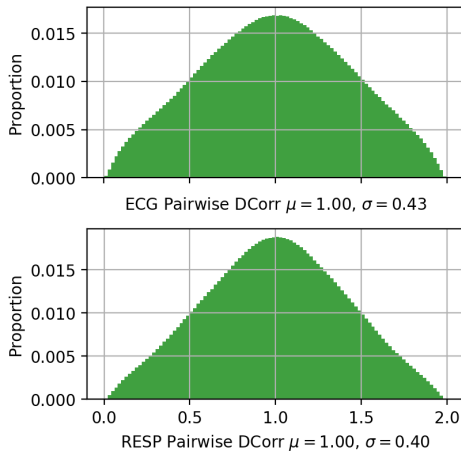


Fig. 2. Distribution of distance correlations between randomly selected pairs of 2-second segments of ECG waveforms, and pairs of 2-second segments of RESP waveforms.

$$DCorr(x, y) = 1 - \frac{(x - \bar{x}) \cdot (y - \bar{y})}{\|x - \bar{x}\|_2 \|y - \bar{y}\|_2} \quad (1)$$

To evaluate the reconstructed waveform against the original input, we use the *Distance Correlation*, or DCorr, between the two sequences, as defined in Equation 1 [20]. The distance correlation ranges from 0 when the two sequences are perfectly correlated, to 2 when they are perfectly anti-correlated, and 1 when there is no correlation. Distance correlation is continuous and differentiable, allowing its direct utilization as a neural network loss, and more heavily considers errors in extreme values than other loss functions like mean square error. Figure 2 shows that the distance correlation between randomly selected 2-second segments of the ECG or RESP waveforms is relatively symmetric, with the majority of samples having little or no correlation. This is primarily because the segments are sampled at arbitrary phase, i.e. the heartbeats for two segments may not line up even when the heart rates are similar. We therefore hypothesize that the network's hidden state must encode information about the relative phase of the input signal.

III. COMPRESSING AUTOENCODER

To assess the performance of the autoencoder network, we train the network on the 80% training set for 10 epochs, and select the parameters for the epoch with the best average performance on the validation set. Each physiological waveform is trained independently, so that $N_C = 1$. The sample frequency is 125 samples per second over a 2-second interval, so that $L = 250$. We use $R = 5$ convolutional encoding layers, so that $\lfloor 250 \cdot 2^{-5} \rfloor = 7$, and consider a range of $C = 0, 1, \dots, 5$ compression layers. This results in respective compression rates of $250 : (224, 112, 56, 28, 14, 7)$. The network is trained using an Adam as the standard deep learning gradient descent optimization algorithm [21] with a learning rate of 0.01 and a batch size of 64. All computation and deep learning was performed with Python and Pytorch [22] running on an NVIDIA Tesla M60 GPU, and each network trained in under 10 hours.

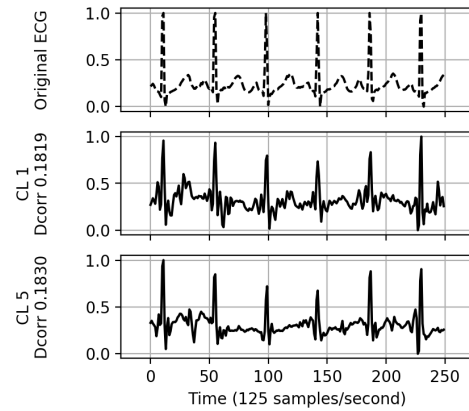


Fig. 3. Reconstructed ECG waveforms across a range of compression layers.

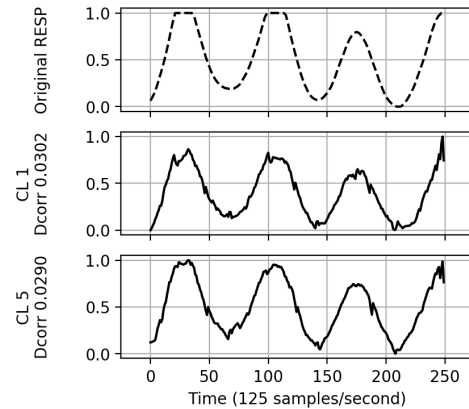


Fig. 4. Reconstructed RESP waveforms across a range of compression layers.

Figures 3 and 4 visually demonstrate how the ECG and RESP autoencoder performances vary according to the compression level. The best average distance correlation across 10 training epochs for both the ECG and RESP autoencoders is shown for a range of compression layers in Figure 5. We can see that due to the information loss in the hidden state, the

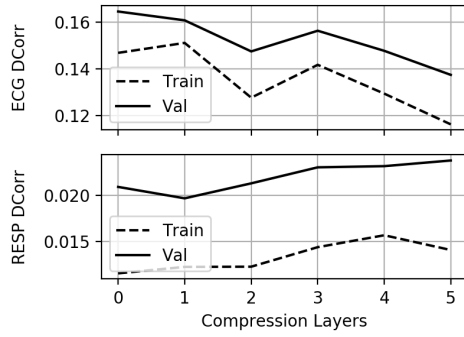


Fig. 5. Autoencoder performance versus the number of compression layers.

autoencoder introduces a significant amount of noise into the reconstructed waveform. In the following section, we consider how relaxing the compression requirement can result in a much more powerful autoencoder.

IV. MISSING DATA IMPUTATION

Once an autoencoder has been trained, it can be used in a variety of ways to impute missing data. For dense, potentially periodic time-series data like these physiological waveforms, basic interpolation such as linear or spline methods are insufficient, particularly in their inability to handle arbitrary phase shifts. If we wish to impute a missing waveform segment within a longer sequence, the estimated segment should retain the temporal patterns of the non-missing data around it. For example, an estimated ECG should contain QRS complexes of the same shape and heart-rate as the ECG signal that precedes and follows the missing data. If the shape or frequency differs between the preceding and following signals, the estimated waveform should provide a logical and smooth transition.

Since data compression is no longer a primary concern, we construct a more powerful autoencoder with significantly more parameters. To do so, we apply a channel expansion multiplier $K = 32$ to the first convolutional encoding layer, such that each convolutional layer has $N_C \cdot K \cdot 2^R$ channels. We train the expanded autoencoder with 5 convolutional layers, and 5 compression layers, such that the hidden state has shape 32×7 , resulting in the performance shown in Table I.

TABLE I. Expanded autoencoder performance

	Train	Validation
ECG	0.049	0.022
RESP	0.034	0.017

Encoding the preceding and following signals in the compressed hidden space defined by the autoencoder network yields a representation which consolidates all temporal patterns. As a simple example, we consider the imputation model shown in Figure 8. This model takes a 6-second sequence of the physiological waveform, and splits it into three 2-second-long segments. We encode the first and last segments using the convolution and compression layers of the autoencoder, and simply average the two hidden states to obtain an estimated hidden state for the middle segment. We then reconstruct

an estimated middle segment using the decompression and deconvolutional layers of the autoencoder. Figures 6 and 7 show example reconstructions for each independent waveform.

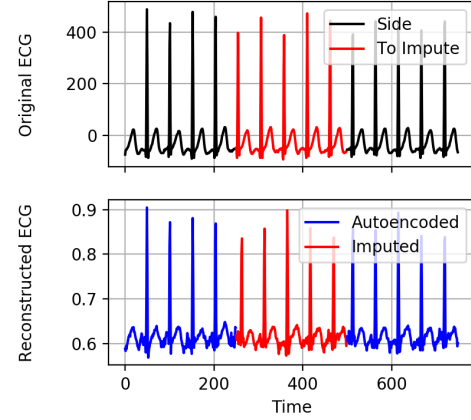


Fig. 6. Autoencoder-based missing data reconstruction. The upper figure shows the original ECG waveform. The lower side channels were compressed to 7 dimensions and reconstructed using the autoencoder. The hidden state of the middle segment was estimated by averaging the hidden states of the side segments, and then reconstructed using the autoencoder deconvolution layers.

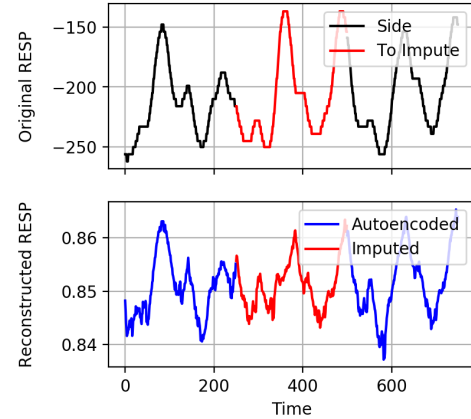


Fig. 7. Autoencoder-based missing data reconstruction. The upper figure shows the original RESP waveform. The lower side channels were compressed to 7 dimensions and reconstructed using the autoencoder. The hidden state of the middle segment was estimated by averaging the hidden states of the side segments, and then reconstructed using the autoencoder deconvolution layers.

V. HIDDEN STATE EXAMINATION

Extracting the compressed (hidden) state of the autoencoder network can be regarded as a fully unsupervised feature extraction method. In this way, the autoencoder acts not only as a compression algorithm for storage, but also as a feature extractor for downstream supervised learning applications. One of the primary advantages of this method is its independence of the specific waveform type. In contrast, a traditional machine learning feature extraction algorithm for the ECG waveform would involve developing specific signal processing algorithms to extract the heart rate, each QRS component duration and magnitude, and statistical summaries of these features' evolution over time. Furthermore, the autoencoder

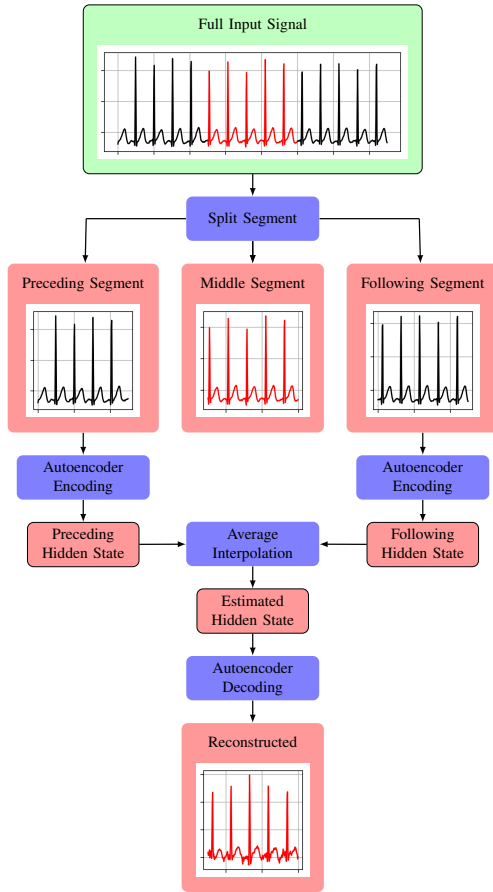


Fig. 8. Missing Data Interpolation using pre-trained network. Each individual segments are compressed to hidden states. The hidden state of the middle segment is estimated as the average of the preceding and following hidden states. This state is passed through the autoencoder's decompression and deconvolutional layers to reconstruct the time series.

network may be easily generalized to handle variable-length inputs, compressing them all into the same contextual space.

While the pure compression ability of the CNN is guaranteed to be competitive with more traditional compression methods, it has the advantage of being specifically tuned to the given loss function, in this case distance correlation. Therefore, the “noise” that is removed is those parts of the signal which are not relevant to the autoencoder reconstruction. Furthermore, training the network on a different loss function, such as a predictive loss for a classification network, will adapt the compression focus accordingly. Examination of the compressed states of the network can provide insights regarding the overall dataset structure and underlying associations between waveform samples. In this section, we analyze compressed states which were generated using $R = 5$ convolutional layers, and $C = 5$ compression layers for a 250:7 compression ratio.

$$\mathcal{L} = \sum_{i=1}^N \|x^i - \mu_{c^i}\|^2 \quad (2)$$

We hypothesize that the compressed waveform states may be separated into groups, each indicative of a particular type of waveform. We therefore perform a K-means clustering on

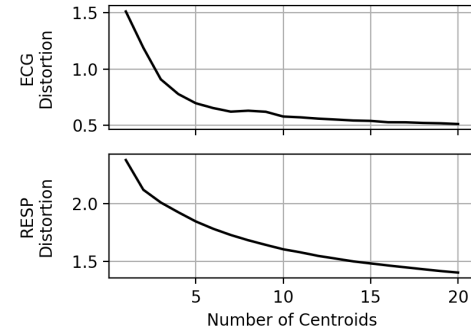


Fig. 9. K-means distortion on the 7-dimensional hidden states versus the number of centroids.

the 7-dimensional hidden states. Figure 9 shows the average distortion for each number of cluster centroids, or number of waveform types. The average distortion is defined in Equation 2, where x^i is the i 'th sample, and c^i is the closest centroid to that sample. We can see that for the compressed ECG samples, the majority of the distortion reduction comes within the first 4 cluster centroids, whereas for the RESP samples, the distortion reduction is more gradual.

Although these cluster are defined in the full 7-dimensional hidden space, for visualization purposes, we show the first two principal components of the hidden states in Figures 10 and 11. Each sample is colored according to its index among a 4-cluster K-means classification. We can see that while the ECG may be logically separated into multiple waveform types, the RESP waveform is less quantifiably separable. We further consider the joint sample distribution of ECG and RESP hidden state classes in Table II.

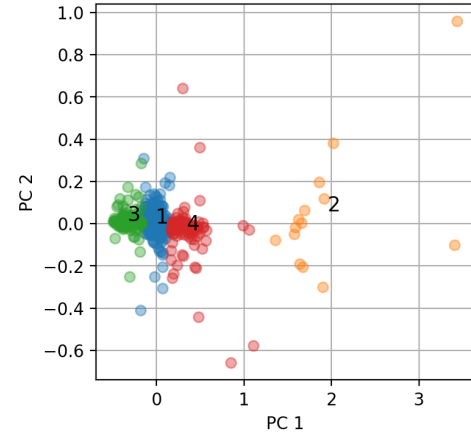


Fig. 10. 2-dimensional Principal Component Analysis (PCA) reduction of the ECG waveform hidden states. 1000 randomly selected samples are shown, colored by their associated cluster.

A. Applications

One application for the joint distribution over compressed states is a further adaptation of the autoencoder for missing data imputation. We first consider a variational autoencoder, for which we have determined the hidden state distribution. For

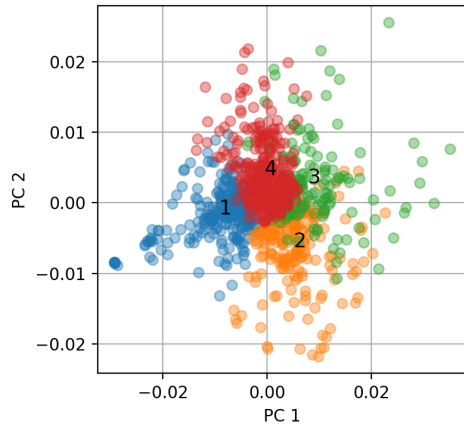


Fig. 11. 2-dimensional Principal Component Analysis (PCA) reduction of the RESP waveform hidden states. 1000 randomly selected samples are shown, colored by their associated cluster.

TABLE II. Joint RESP/ECG hidden state cluster co-occurrence matrix over 10,000 randomly selected samples.

	RESP 1	RESP 2	RESP 3	RESP 4	Total
ECG 1	36	31	31	11	109
ECG 2	961	511	605	242	2319
ECG 3	462	300	334	193	1289
ECG 4	2701	1108	1858	616	6283
Total	4160	1950	2828	1062	10000

each waveform type, sampling a random hidden state from this distribution and reconstructing it using the decompression and deconvolution network layers allows us to generate random waveform samples. Further considering the distribution of hidden states in each cluster, or class, could allow us to generate random waveforms of the corresponding class. Finally, if missing data from a single channel, but possessing a separate channel for the same time interval, we can consider the joint distribution to determine the most likely class for the missing waveform, and generate an approximated replacement.

VI. CONCLUSIONS AND FUTURE WORK

Unsupervised deep learning models can enable a wide range of analyses, dataset completion algorithms, pre-processing, and feature extraction or dimension reduction methods for physiological data. These methods may be adapted as a preliminary stage to enable an even wider range of supervised learning models, with the ultimate goal of improving real-time automated patient diagnostics. The most significant advantage of these models is their extremely generalizable structure. With only minor alterations to the network, these models can support nearly any type of physiological time-series data, multiple channel decomposition and reconstruction, early-warning prediction, or long-term patient trend analysis.

ACKNOWLEDGMENTS

Daniel Miller is supported by Stanford University through the Stanford Graduate Fellowship. Andrew Ward is supported by the Department of Defense through the National Defense Science & Engineering Graduate Fellowship.

REFERENCES

- [1] P. Rajpurkar, A. Y. Hannun, M. Haghighpanahi, C. Bourn, and A. Y. Ng, "Cardiologist-level arrhythmia detection with convolutional neural networks," *arXiv preprint arXiv:1707.01836*, 2017.
- [2] Q. Li, C. Rajagopalan, and G. D. Clifford, "A machine learning approach to multi-level ecg signal quality classification," *Computer methods and programs in biomedicine*, vol. 117, no. 3, pp. 435–447, 2014.
- [3] A. H. Shoen and J. V. Guttag, "Application of machine learning to epileptic seizure detection," in *Proceedings of the 27th International Conference on Machine Learning (ICML-10)*, 2010, pp. 975–982.
- [4] D. Miller, D. Scheinker, and N. Bambos, "A practical approach to machine learning for clinical decision support," in *International Conference on Health Care Systems Engineering*. Springer, 2017, pp. 111–120.
- [5] M. L. Brown and J. F. Kros, "The impact of missing data on data mining," *Data mining: Opportunities and challenges*, vol. 1, pp. 174–198, 2003.
- [6] M. Hejazi, S. A. R. Al-Haddad, Y. P. Singh, S. J. Hashim, and A. F. A. Aziz, "Multiclass support vector machines for classification of ecg data with missing values," *Applied Artificial Intelligence*, vol. 29, no. 7, pp. 660–674, 2015.
- [7] T. Quinn, S. Johnsen, C. P. Gale, H. Snooks, S. McLean, M. Woollard, C. Weston, M. I. N. A. P. M. S. Group *et al.*, "Effects of prehospital 12-lead ecg on processes of care and mortality in acute coronary syndrome: a linked cohort study from the myocardial ischaemia national audit project," *Heart*, pp. heartjnl-2013, 2014.
- [8] B. A. Cattle, P. D. Baxter, D. C. Greenwood, C. P. Gale, and R. M. West, "Multiple imputation for completion of a national clinical audit dataset," *Statistics in medicine*, vol. 30, no. 22, pp. 2736–2753, 2011.
- [9] G. D. Clifford, F. Azuaje, and P. McSharry, "Ecg statistics, noise, artifacts, and missing data," *Advanced Methods and Tools for ECG Data Analysis*, vol. 6, p. 18, 2006.
- [10] F. Bashir and H.-L. Wei, "Handling missing data in multivariate time series using a vector autoregressive model based imputation (var-im) algorithm: Part i: Var-im algorithm versus traditional methods," in *Control and Automation (MED), 2016 24th Mediterranean Conference on*. IEEE, 2016, pp. 611–616.
- [11] H. S. Anderson and M. R. Gupta, "Expected kernel for missing features in support vector machines," in *Statistical signal processing workshop (SSP), 2011 IEEE*. IEEE, 2011, pp. 285–288.
- [12] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," *arXiv preprint arXiv:1506.00327*, 2015.
- [13] P. Bashivan, I. Rish, M. Yeasin, and N. Codella, "Learning representations from eeg with deep recurrent-convolutional neural networks," *arXiv preprint arXiv:1511.06448*, 2015.
- [14] A. v. d. Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, "Wavenet: A generative model for raw audio," *arXiv preprint arXiv:1609.03499*, 2016.
- [15] B. Bollepalli, L. Juvela, P. Alku *et al.*, "Generative adversarial network-based glottal waveform model for statistical parametric speech synthesis," in *Proc. INTERSPEECH*, 2017, pp. 3394–3398.
- [16] L. Juvela, B. Bollepalli, X. Wang, H. Kameoka, M. Airaksinen, J. Yamagishi, and P. Alku, "Speech waveform synthesis from mfcc sequences with generative adversarial networks," *arXiv preprint arXiv:1804.00920*, 2018.
- [17] S. Takaki and J. Yamagishi, "A deep auto-encoder based low-dimensional feature extraction from fft spectral envelopes for statistical parametric speech synthesis," in *Acoustics, Speech and Signal Processing (ICASSP), 2016 IEEE International Conference on*. IEEE, 2016, pp. 5535–5539.
- [18] X. Lu, Y. Tsao, S. Matsuda, and C. Hori, "Speech enhancement based on deep denoising autoencoder," in *Interspeech*, 2013, pp. 436–440.
- [19] S. Pascual, A. Bonafonte, and J. Serra, "Segan: Speech enhancement generative adversarial network," *arXiv preprint arXiv:1703.09452*, 2017.
- [20] G. J. Székely, M. L. Rizzo, and N. K. Bakirov, "Measuring and testing dependence by correlation of distances," *The annals of statistics*, pp. 2769–2794, 2007.
- [21] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [22] A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, "Automatic differentiation in pytorch," 2017.